

RetroDiff: Training Objective Derivation

Let x denote a clean data sample, x_t its noised version at diffusion step t , q the fixed forward noising distribution, and p_θ the learned reverse distribution parameterised by θ . Starting from the log-likelihood, Jensen’s inequality gives the *evidence lower bound* (ELBO):

$$\log p_\theta(x) \geq \mathbb{E}_{q(x_{1:T}|x)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x)} \right] = -\mathcal{L}_{\text{ELBO}} \quad (1)$$

Maximising the right-hand side of (1) is equivalent to minimising $\mathcal{L}_{\text{ELBO}}$. Expanding the joint distributions and applying the Markov structure of the forward process yields the canonical decomposition into a reconstruction term and $T-1$ per-step KL divergences:

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_q[-\log p_\theta(x_0 | x_1)]}_{\text{reconstruction}} + \sum_{t=2}^T \underbrace{\text{KL}(q(x_{t-1} | x_t, x) \parallel p_\theta(x_{t-1} | x_t))}_{\text{step-}t \text{ denoising}} \quad (2)$$

Each KL term in (2) has a tractable Gaussian form because the forward posterior $q(x_{t-1}|x_t, x)$ is Gaussian. **RetroDiff**’s key contribution is to condition the objective on a set $R(x)$ of k retrieved exemplars drawn from the training corpus at inference time, leading to the retrieval-augmented objective:

$$\mathcal{L}_{\text{RetroDiff}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\mathcal{L}_{\text{ELBO}}(x | R(x))] \quad (3)$$

Minimising (3) trains the reverse process to exploit retrieved neighbours, enabling sharper reconstructions without architectural changes.